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Project: Titanic Missing Values Handling and Data Analysis

Objective

- The objective of this project is to perform data exploration, handle missing values, clean the dataset, visualize passenger information, and derive meaningful insights from the Titanic dataset.

1. Import Required Libraries

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
```

2. Load the Dataset

```
df = pd.read_csv("titanic.csv")
```

3. Dataset Overview

3.1 First 5 records

```
df.head()
```

```
   PassengerId  Survived  Pclass  \
0            1         0       3
1            2         1       1
2            3         1       3
3            4         1       1
4            5         0       3

                                                Name     Sex   Age  SibSp  \
0                            Braund, Mr. Owen Harris    male  22.0      1
1  Cumings, Mrs. John Bradley (Florence Briggs Th...  female  38.0      1
2                             Heikkinen, Miss. Laina  female  26.0      0
3       Futrelle, Mrs. Jacques Heath (Lily May Peel)  female  35.0      1
4                           Allen, Mr. William Henry    male  35.0      0

   Parch            Ticket     Fare Cabin Embarked
0      0         A/5 21171   7.2500   NaN        S
1      0          PC 17599  71.2833   C85        C
2      0  STON/O2. 3101282   7.9250   NaN        S
3      0            113803  53.1000  C123        S
4      0            373450   8.0500   NaN        S
```

3.2 Dataset Shape

```
df.shape
```

```
(891, 12)
```

3.3 Dataset Information

```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 12 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
```

```
 0   PassengerId  891 non-null    int64
 1   Survived     891 non-null    int64
 2   Pclass       891 non-null    int64
 3   Name         891 non-null    str
 4   Sex          891 non-null    str
 5   Age          714 non-null    float64
 6   SibSp        891 non-null    int64
 7   Parch        891 non-null    int64
 8   Ticket       891 non-null    str
 9   Fare         891 non-null    float64
 10  Cabin        204 non-null    str
 11  Embarked     889 non-null    str
dtypes: float64(2), int64(5), str(5)
memory usage: 83.7 KB
```

3.4 Column Names

```
df.columns
```

```
Index(['PassengerId', 'Survived', 'Pclass', 'Name', 'Sex', 'Age', 'SibSp',
       'Parch', 'Ticket', 'Fare', 'Cabin', 'Embarked'],
      dtype='str')
```

3.5 Statistical Summary

```
df.describe()
```

```
       PassengerId    Survived      Pclass         Age       SibSp  \
count   891.000000  891.000000  891.000000  714.000000  891.000000
mean    446.000000    0.383838    2.308642   29.699118    0.523008
std     257.353842    0.486592    0.836071   14.526497    1.102743
min       1.000000    0.000000    1.000000    0.420000    0.000000
25%     223.500000    0.000000    2.000000   20.125000    0.000000
50%     446.000000    0.000000    3.000000   28.000000    0.000000
75%     668.500000    1.000000    3.000000   38.000000    1.000000
max     891.000000    1.000000    3.000000   80.000000    8.000000

            Parch        Fare
count  891.000000  891.000000
mean     0.381594   32.204208
std      0.806057   49.693429
min      0.000000    0.000000
25%      0.000000    7.910400
50%      0.000000   14.454200
75%      0.000000   31.000000
max      6.000000  512.329200
```

4. Null Value Analysis

4.1 Count Missing Values

```
df.isnull().sum()
```

```
PassengerId      0
Survived         0
Pclass           0
Name             0
Sex              0
Age            177
SibSp            0
Parch            0
Ticket           0
```

```
Fare             0
Cabin          687
Embarked         2
dtype: int64
```

4.2 Percentage of Missing Values

```
missing_percentage = (df.isnull().sum()/len(df)) * 100
missing_percentage
```

```
PassengerId     0.000000
Survived        0.000000
Pclass          0.000000
Name            0.000000
Sex             0.000000
Age            19.865320
SibSp           0.000000
Parch           0.000000
Ticket          0.000000
Fare            0.000000
Cabin          77.104377
Embarked        0.224467
dtype: float64
```

5. Data Cleaning

5.1 Handling Missing Values in Age

```
df["Age"] = df["Age"].fillna(df["Age"].mean())
```

5.2 Handling Missing Values in Embarked

```
df["Embarked"] = df["Embarked"].fillna(df["Embarked"].mode()[0])
```

5.3 Dropping Cabin Column

```
df.drop("Cabin", axis=1, inplace=True)
```

5.4 Verification After Cleaning

```
df.isnull().sum()
```

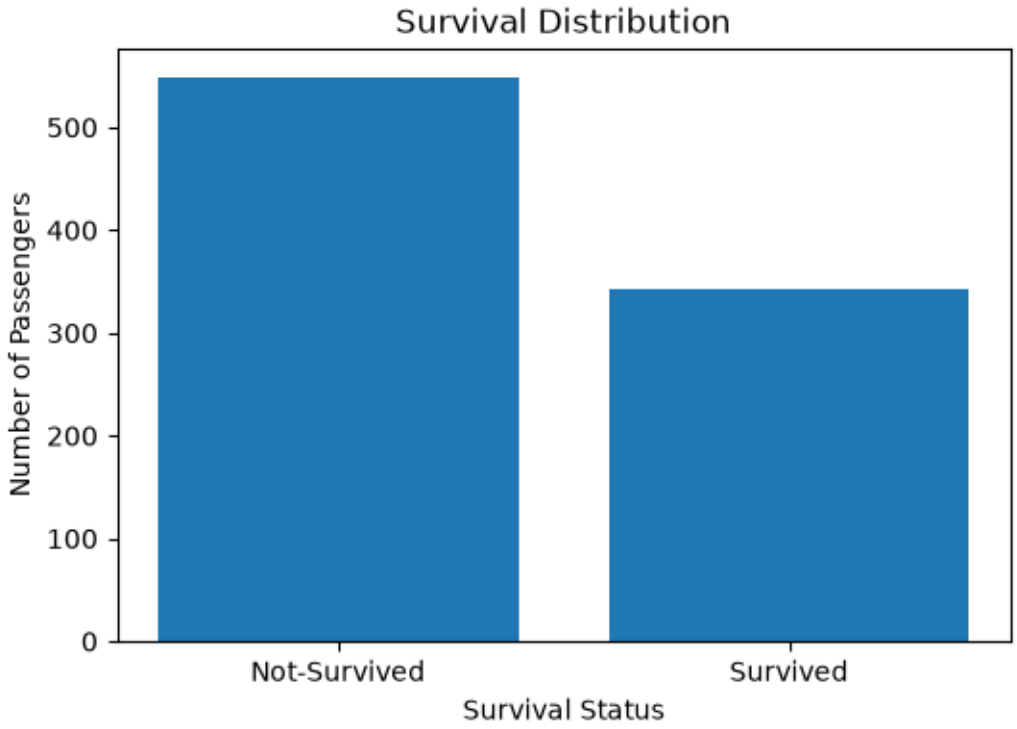
```
PassengerId    0
Survived       0
Pclass         0
Name           0
Sex            0
Age            0
SibSp          0
Parch          0
Ticket         0
Fare           0
Embarked       0
dtype: int64
```

6. Data Visualization

6.1 Survival Distribution

```
survival = df["Survived"].value_counts()
plt.figure(figsize=(6,4))
plt.title("Survival Distribution")
```

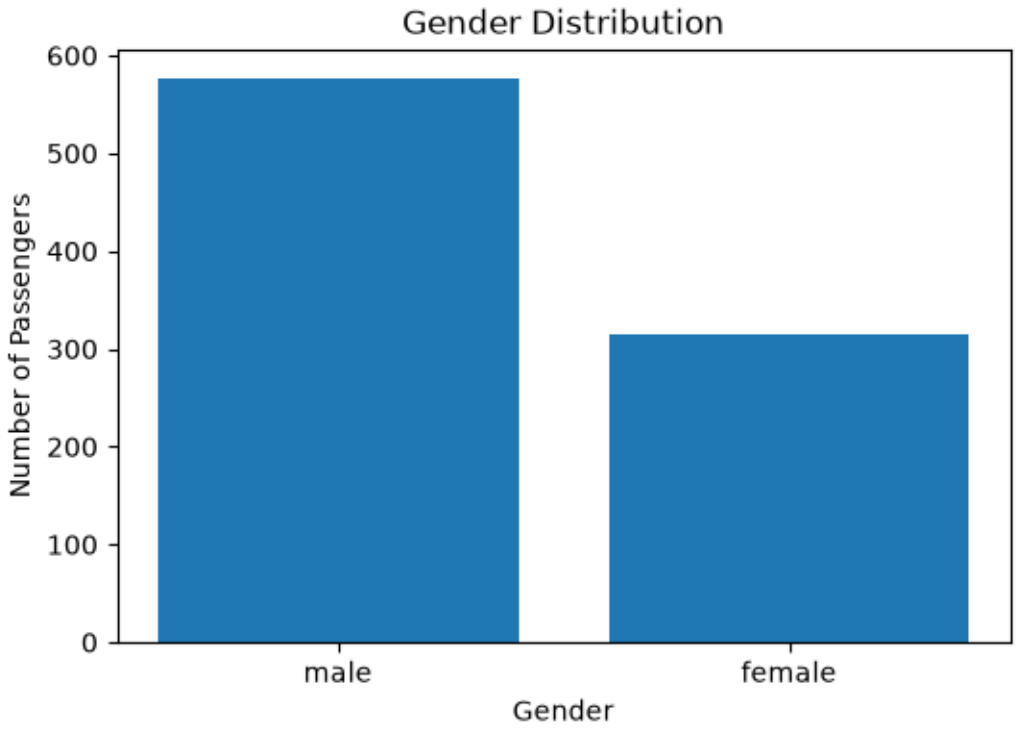
```
plt.xlabel("Survival Status")
plt.ylabel("Number of Passengers")
plt.bar(["Not-Survived", "Survived"], survival)
plt.show()
```



Most passengers did not survive

6.2 Gender Distribution

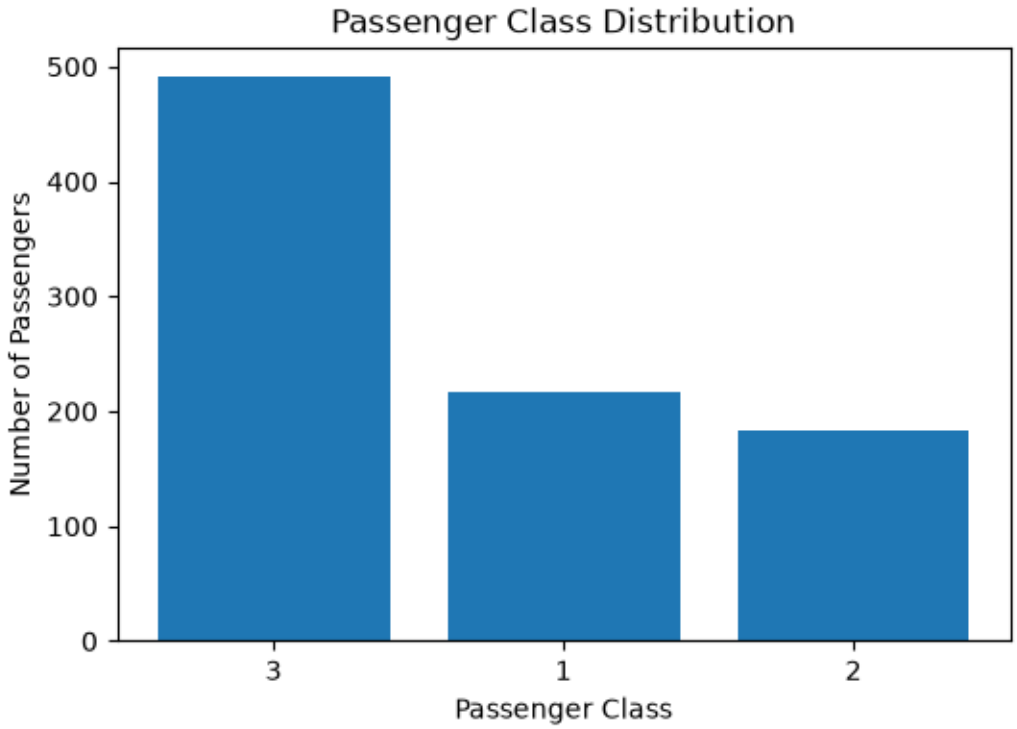
```
gender = df["Sex"].value_counts()
plt.figure(figsize=(6,4))
plt.title("Gender Distribution")
plt.xlabel("Gender")
plt.ylabel("Number of Passengers")
plt.bar(gender.index, gender.values)
plt.show()
```



Male passengers were more than female passengers

6.3 Passenger Class Distribution

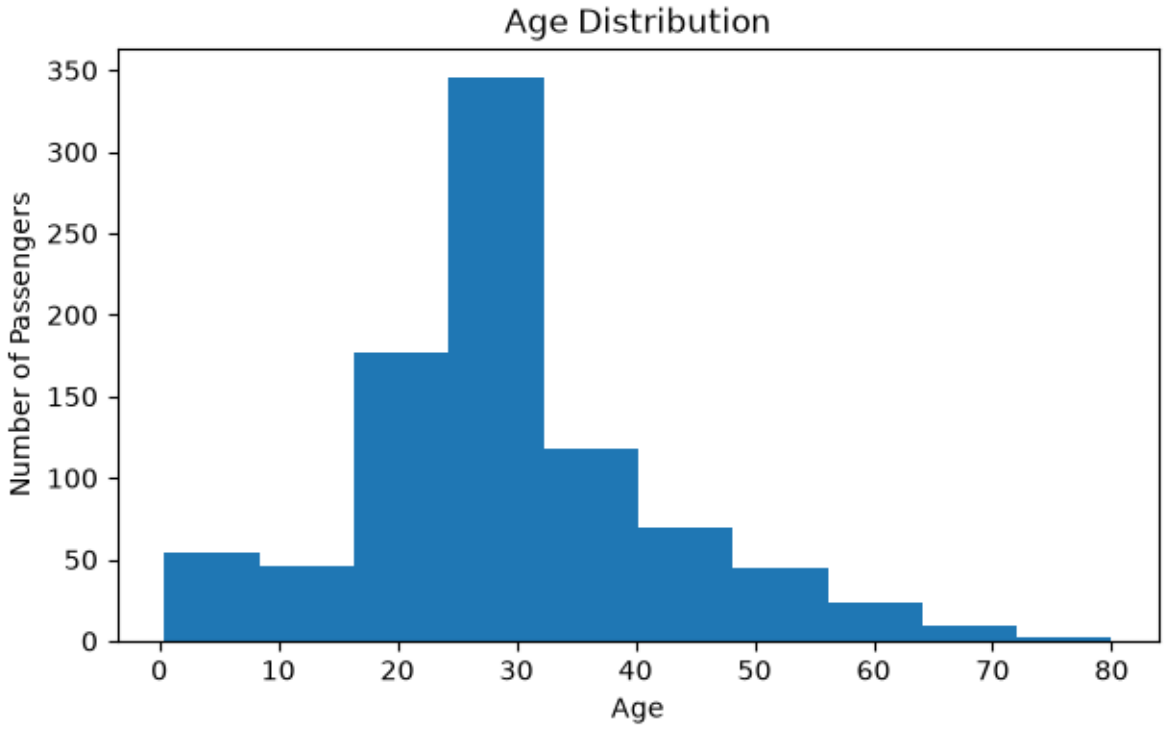
```
pclass = df["Pclass"].value_counts()
plt.figure(figsize=(6,4))
plt.title("Passenger Class Distribution")
plt.xlabel("Passenger Class")
plt.ylabel("Number of Passengers")
plt.bar(pclass.index.astype(str), pclass.values)
plt.show()
```



Most of the passengers travelled in Third Class

6.4 Age Distribution

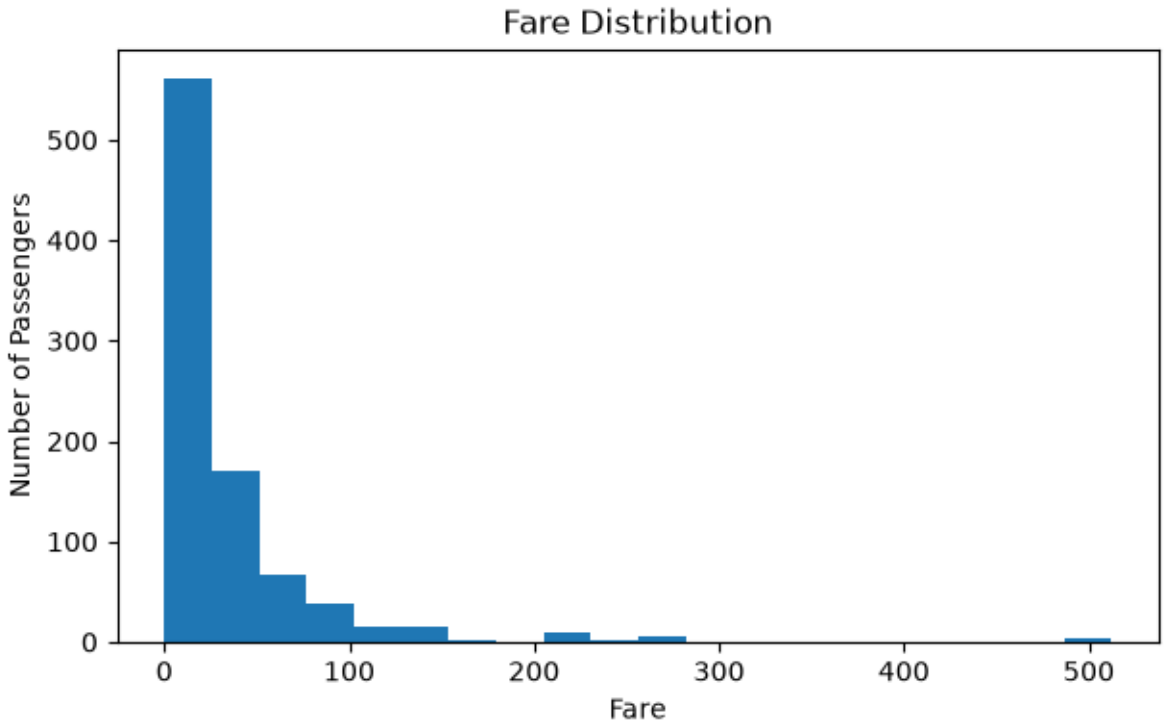
```
plt.figure(figsize=(7,4))
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Number of Passengers")
plt.hist(df["Age"], bins=10)
plt.show()
```



Most passengers were Young Adults

6.5 Fare Distribution

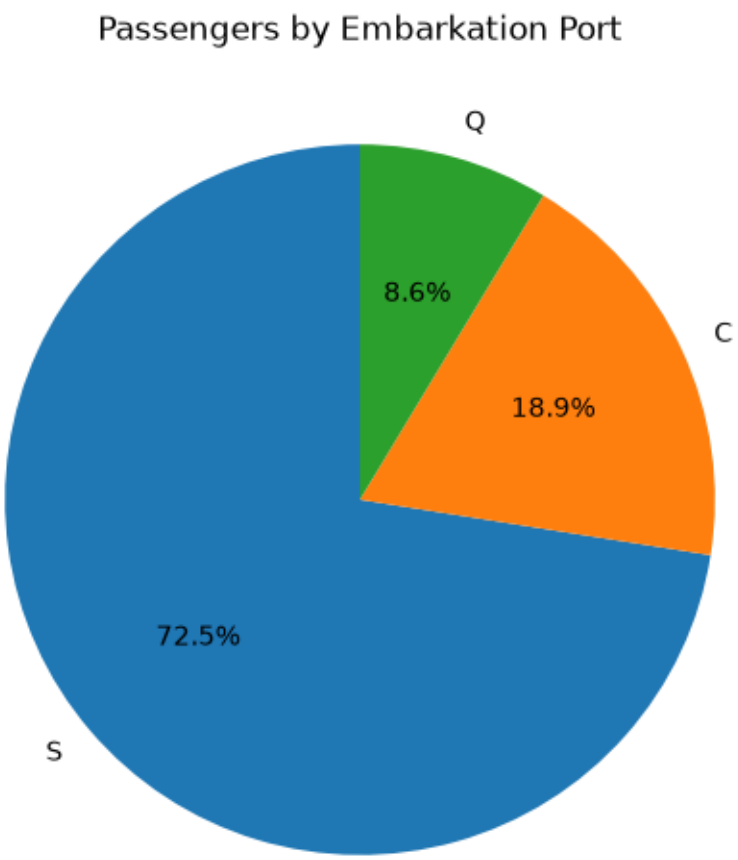
```
plt.figure(figsize=(7,4))
plt.title("Fare Distribution")
plt.xlabel("Fare")
plt.ylabel("Number of Passengers")
plt.hist(df["Fare"], bins=20)
plt.show()
```



Most passengers paid lower fares, very few paid high fares

6.6 Embarked Plot

```
embarked = df["Embarked"].value_counts()
plt.figure(figsize=(6,6))
plt.title("Passengers by Embarkation Port")
plt.pie(embarked.values, labels=embarked.index, autopct="%1.1f%%", startangle=90)
plt.show()
```



Most of the passengers boarded from Southampton (S)

7. Statistical Analysis

```
print("Average Age: ",df["Age"].mean())
print("Average Fare: ",df["Fare"].mean())
print("Maximum Fare:" ,df["Fare"].max())
print("Minimum Fare:" ,df["Fare"].min())
```

```
Average Age:  29.69911764705882
Average Fare:  32.204207968574636
Maximum Fare: 512.3292
Minimum Fare: 0.0
```

8. Observations

- The dataset contains 891 passenger records.
- Age contained 177 missing values.
- Cabin contained 687 missing values.
- Embarked contained 2 missing values.
- Most passengers traveled in third class.

9. Conclusion

- This project successfully explored the Titanic dataset by performing data exploration, identifying and handling missing values, cleaning the dataset, and visualizing passenger information. Basic statistical analysis provided useful insights into passenger demographics and survival patterns. The cleaned dataset can now be used for further analysis or machine learning tasks.